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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/31/2006 |
| **PI Name:** | Vazquez, Marcelo Ph.D. | | |
| **Project Title:** | CNS Damage and Countermeasures | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Radiation Effects Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | GOVERNMENT | **Phone:** | (631) 344-3443 |
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| **City:** | Upton | **State:** | NY |
| **Zip Code:** | 11973 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 12/01/2001 | **End Date:** | 04/30/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Space travel beyond the Earth's protective magnetic field (for example, to Mars) will involve exposure of astronauts to irradiation by high-energy nuclei such as 56Fe (HZE radiation), which are a component of galactic cosmic rays. These particles have high linear energy transfer (LET) and are expected to irreversibly damage cells they traverse. Exposure to HZE radiation may therefore cause progressive deterioration of brain function, adding to other inescapable damage involved in normal aging. We propose a study of the hypothesis that long-term behavioral alterations are induced after exposure of the brain to 1 GeV/n iron particles with fluences of 1 to 8 particles/cell targets. Previous studies support this notion but are not definitive, especially with regard to long-term effects. Our principal goal is to examine the neurological effects of high-LET radiation on C57BL/6 mice using a series of behavioral tests to unveil the temporal expression of altered behaviors in the radiation response, as well as the means, which can modulate these responses. The studies proposed in this application are designed to: 1) Characterize the behavioral consequences after exposure to low-fluences of heavy ions and protons on C57BL/6 mice. The main behavioral endpoints to be used in these studies are locomotor | | |

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| | activity to evaluate the integrity of striatal dopaminergic pathways, and spatial reference memory to probe hippocampal cholinergic pathways. 2) Characterize the neurochemical and structural changes induced by heavy ions and protons. 3) To develop countermeasures to protect neural cell populations exposed to low fluences of heavy ions and protons. The project will test methods to protect injured neural cells based on their molecular and cellular mechanisms that may regulate neural cell survival in the central nervous system. Among the methods that will be studied is the direct administration of neuroprotective molecules as well as the modulation of apoptotic pathways by pharmacological manipulation. The effects of 3 different neuro/radioprotectors (GM1, melatonin and PTF-?) on the levels of radiation induced neurochemical and structural damage will be compared with the level of behavioral alterations to determine a cause/effect relationship. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | . |
| **Task Progress:** | No progress report this period. |
| **Bibliography Type:** | Description: (Last Updated: ) |